

ANALYSTLAB AFRICA

HealthConnect Clinic Experience Lab — Week 4

Data Analytics Track — Initial Analysis Document

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1. Dataset Overview

The HealthConnect Appointment Dataset contains 5,000 fictional, anonymized appointment records across 18 fields, confirmed against HealthConnect_Data_Dictionary.pdf. The dataset covers:

- Identifiers: appointment_id (unique, no duplicates), patient_id (repeats across multiple appointments per patient)
- Demographics: gender (Female, Male, Prefer not to say), age (18-80), age_group (6 bands, 18-24 through 65+)
- Appointment details: appointment_type (4 categories), booking_date, appointment_date, appointment_day, appointment_time (Morning/Afternoon/Evening)
- Booking behaviour: booking_lead_days (days between booking and appointment)
- History: previous_appointments, previous_no_shows (per patient, cumulative)
- Reminders: reminder_sent (Yes/No), reminder_channel (SMS, WhatsApp, Email, or blank when no reminder was sent)
- Clinic logistics: distance_to_clinic_km, waiting_time_minutes
- Outcome: appointment_outcome (Attended, No-Show, Cancelled) — the field this track's analysis centres on

The dataset is well-suited to this track's descriptive/diagnostic focus, and the same appointment_outcome field can support the Data Science track's predictive modelling of no-shows.

2. Data Quality Assessment

A full data quality review was carried out directly on HealthConnect_Appointment_Data.csv (5,000 rows).

Summary of findings:

Check	Result	Action Taken
Duplicate appointment_id / full-row duplicates	0 found	None needed
Missing values — reminder_channel	1,366 blank (27.3%) — all correspond exactly to rows where reminder_sent = No	Filled with “None” in cleaned file (logically expected, not a data error)
Missing values — distance_to_clinic_km	90 blank (1.8%)	Left as-is; minor enough not to bias analysis, to be excluded row-wise in distance-specific views
Missing values — waiting_time_minutes	60 blank (1.2%)	Left as-is; same treatment as above
Logical check: previous_no_shows > previous_appointments	0 violations	None needed
Logical check: negative booking_lead_days	0 violations	None needed
Logical check: appointment_date before booking_date	0 violations	None needed

Category consistency (gender, outcome, etc.)	Clean, consistent labelling throughout — no casing or duplicate-category issues	None needed
Date field types	booking_date and appointment_date stored as text in the raw file	Converted to proper date type in cleaned file

Overall, the dataset is in strong condition — no duplicates, no logical inconsistencies, and only minor, explainable gaps in three columns. A cleaned version has been saved separately as HealthConnect_Appointment_Data_Cleaned.csv, with dates converted to proper date type and reminder_channel blanks filled with “None”; the original file was not modified.

3. Relevant Business Questions

- What proportion of scheduled appointments end in a no-show? (Answer: 48.5% of all 5,000 appointments — nearly 1 in 2)
- Which patient or appointment characteristics are most associated with missed appointments?
- Does receiving a reminder measurably relate to a lower no-show rate?
- Are there particular days or time slots with disproportionately high no-show rates?
- How much appointment capacity is lost to no-shows?

4. Potential KPIs

The following KPIs are proposed, each linked to a business question above. Initial figures from the raw dataset are included to demonstrate feasibility — deeper analysis and visualisation will follow in Week 5.

KPI	Definition	Linked Question	Initial Figure
No-Show Rate (%)	No-shows ÷ total scheduled appointments	Q1	48.5%
No-Show Rate by Lead Time Band	No-show rate for Same-day, 1–7 days, and 8+ days lead time	Q2	24.7% / 28.3% / 51.5%
Reminder Effectiveness Rate	No-show rate with vs. without a reminder sent	Q3	47.4% vs 51.4%
No-Show Rate by Day/Time Slot	No-show rate by appointment_day and appointment_time	Q4	Range 46.6–50.5%
Lost Appointment Slots	Count of no-show appointments (unused slots)	Q5	2,423 of 5,000

Two early signals stand out and merit deeper investigation in Week 5: (1) booking lead time shows a striking gradient — appointments booked 8+ days ahead are roughly twice as likely to be missed as same-day or withinweek bookings, making it the strongest single pattern found so far; and (2) reminders show only a modest ~4 percentage-point difference in no-show rate, weaker than initially expected, which is worth exploring further (e.g. whether reminder channel, not just reminder presence, matters more).

5. Initial Analysis Approach

- Dataset loaded and validated against the data dictionary; data quality checks completed (Section 2)
- Cleaned file saved separately as HealthConnect_Appointment_Data_Cleaned.csv; original file untouched
- Initial exploratory cross-tabulations run for all five proposed KPIs to confirm feasibility (figures shown in Section 4)
- Week 5 will build on this with fuller exploratory analysis (e.g. reminder channel breakdown, age/distance relationships) and initial visualisations

- Excel/Power BI to be used for dashboard-facing exploration; Python (pandas) retained for deeper diagnostic work

6. Assumptions, Limitations, Risks and Dependencies

Assumptions

- appointment_outcome reliably distinguishes No-Show from Cancelled and Attended, and reminder_sent reflects a reminder being sent rather than confirmed as received/read

Limitations

- This is a fictional, synthetic dataset; patterns found (e.g. the lead-time effect) are illustrative for the Experience Lab and should not be read as real clinical or operational findings
- distance_to_clinic_km and waiting_time_minutes each have a small amount of missing data (~1.8% and ~1.2%); any KPI or analysis using these fields will exclude those rows rather than impute values, to avoid introducing bias

Risks and Dependencies

- The strength of the booking_lead_days relationship should be validated further (e.g. controlling for appointment_type) before treating it as the leading driver in later stages
- KPI definitions here should stay coordinated with whatever target variable the Data Science track selects, to keep the shared HealthConnect project consistent across tracks
- Cleaned data handling (separate file, dates converted, reminder_channel filled) should be communicated clearly so other trackmates reusing this file understand what was changed